

HW #3 - Quadratics Forms Review

Date _____ Period _____

Identify the vertex and axis of symmetry of each. Use your graphing calculator to determine the zeros of the function.

1) $y = 4x^2 - 48x + 137$

2) $y = -2x^2 + 32x - 138$

3) $y = -\frac{7}{8}x^2 + \frac{35}{4}x - \frac{239}{8}$

4) $y = 4x^2 + 72x + 324$

Identify the vertex, axis of symmetry, direction of opening, and min/max value of each.

5) $y = 2x^2 + 20x + 45$

6) $y = -4x^2 - 40x - 96$

7) $y = -x^2 + 12x - 39$

8) $y = -2x^2 - 24x - 82$

Identify the vertex, axis of symmetry, and y-intercept of each. Then sketch the graph. Remember to plot at least 5 points per parabola. Determine the domain, range, and interval of increase and decrease.

9) $y = -x^2 - 2x - 4$

10) $y = 2x^2 + 4x - 3$

11) $y = 2(x - 5)^2 - 5$

12) $y = -(x - 3)^2$

$$13) \ y = -(x - 6)(x - 2)$$

$$14) \ y = 2(x + 7)(x + 5)$$

Answers to HW #3 - Quadratics Forms Review

1) Vertex: $(6, -7)$
Axis of Sym.: $x = 6$

2) Vertex: $(8, -10)$
Axis of Sym.: $x = 8$

3) Vertex: $(5, -8)$
Axis of Sym.: $x = 5$

4) Vertex: $(-9, 0)$
Axis of Sym.: $x = -9$

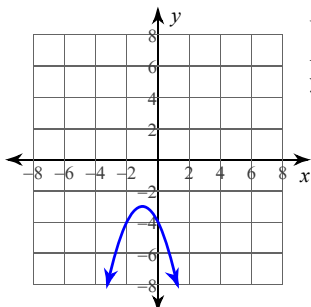
5) Vertex: $(-5, -5)$
Axis of Sym.: $x = -5$
Opens: Up
Min value = -5

6) Vertex: $(-5, 4)$
Axis of Sym.: $x = -5$
Opens: Down
Max value = 4

7) Vertex: $(6, -3)$
Axis of Sym.: $x = 6$
Opens: Down
Max value = -3

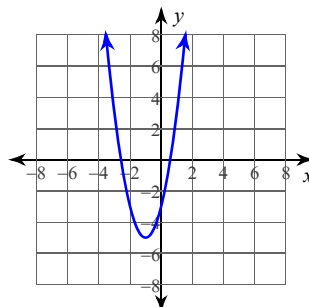
8) Vertex: $(-6, -10)$
Axis of Sym.: $x = -6$
Opens: Down
Max value = -10

9)



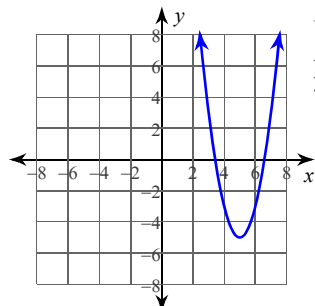
Vertex: $(-1, -3)$
Axis of Sym.: $x = -1$
y-int: -4

10)



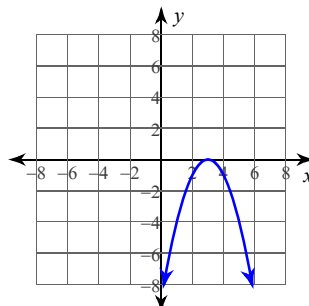
Vertex: $(-1, -5)$
Axis of Sym.: $x = -1$
y-int: -3

11)



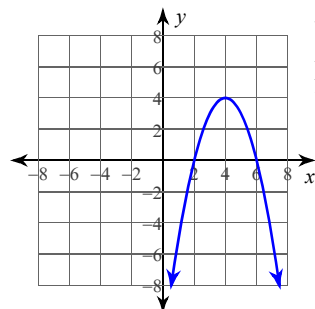
Vertex: $(5, -5)$
Axis of Sym.: $x = 5$
y-int: 45

12)



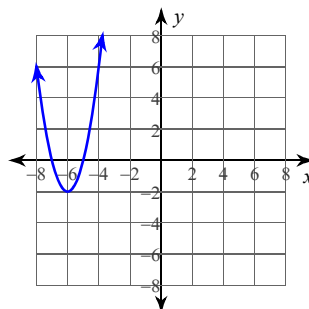
Vertex: $(3, 0)$
Axis of Sym.: $x = 3$
y-int: -9

13)



Vertex: $(4, 4)$
Axis of Sym.: $x = 4$
y-int: -12

14)



Vertex: $(-6, -2)$
Axis of Sym.: $x = -6$
y-int: 70